1 Forward kinematics models

1.1 DH model

The DH frames are assigned from the CuRo6R URDF, with frame $\{0\}$ at `base_link`, joint i rotating about z_{i-1}, and $\boldsymbol{q} = \mathbf{0}$ matching the URDF home configuration. We use the standard DH transform ${}^{i-1}T_i = R_z(\theta_i)\,T_z(d_i)\,T_x(a_i)\,R_x(\alpha_i)$. Table 1 gives the resulting parameters and the geometric choices that determine them. Each twist α_i rotates z_{i-1} onto z_i about $+x_i$ by the right-hand rule. The only fixed joint-angle offset is $\theta_3 - q_3 = \pi/2$. The full sequence of frame drawings is given in Appendix A.

Table 1: One valid DH assignment for CuRo6R. Lengths are in metres; parameter angles are in radians, with degrees used in the explanations.

i	d_i	θ_i	α_i	a_i	Geometric construction at home
1	1	q_1	$-\pi/3$	0.8	Rise $0.5 + 0.5$, then move 0.8 along $x_1 = x_0$ to the shoulder. Its axis z_1 lies 30° above $+y_0$, requiring a -60° turn from z_0 about $+x_1$.
2	1.25	q_2	$+\pi/2$	2	Move 1.25 along z_1, then 2 along x_2 to reach joint 3. Here $x_2 = x_1$ and $z_2 = -y_1$, so the twist is a positive quarter-turn.
3	0.75	$q_3 + \pi/2$	$+\pi/2$	0	Axes 3 and 4 meet 0.75 along z_2, so $a_3 = 0$. Choosing $x_3 = y_2$ gives the $+90^\circ$ joint-angle offset; $y_2 \times z_2 = x_2 = z_3$ gives the positive quarter-turn twist.
4	2.5	q_4	$-\pi/2$	0	The wrist is $1.5 + 1$ along z_3 from O_3; intersecting axes give $a_4 = 0$. With $x_4 = x_3$, reaching $z_4 = y_3$ requires a negative quarter-turn.
5	0	q_5	$+\pi/2$	0	Axes 5 and 6 meet at $O_5 = O_4$, so $d_5 = a_5 = 0$. With $x_5 = x_4$, reaching $z_5 = -y_4$ requires a positive quarter-turn.
6	1.5	q_6	0	0	The endpoint O_6 lies 1.5 along z_5 from the wrist. The last two DH z axes coincide and point the same way, giving $a_6 = \alpha_6 = 0$.

Endpoint alignment. The endpoint E is `link_6`. Its origin coincides with O_6, while $x_E = z_6$, $y_E = x_6$ and $z_E = y_6$. The fixed alignment is therefore appended to the DH chain to recover the URDF endpoint pose:

$$ {}^{0}T_E = \left(\prod_{i=1}^{6} {}^{i-1}T_i \right) {}^{6}T_E, \qquad {}^{6}T_E = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{1} $$

1.2 PoE model

The product of exponentials (PoE) model uses the joint axes at the home configuration $\boldsymbol{q} = \mathbf{0}$. For each joint, $\boldsymbol{\omega}_i$ is its unit axis direction and $\boldsymbol{r}_i$ is its physical origin. All vectors are expressed in `base_link`; the endpoint is `link_6`, as in the DH model.

Axis directions. The rotation axes were identified visually in a URDF view at home. Their positive directions follow the URDF joint convention. Let $\boldsymbol{e}_x = (1,0,0)^\mathsf{T}$, $\boldsymbol{e}_y = (0,1,0)^\mathsf{T}$ and $\boldsymbol{e}_z = (0,0,1)^\mathsf{T}$ denote the coordinate unit vectors. The fixed tilt $\gamma = \pi/6$ gives $\boldsymbol{\omega}_2 = R_x(\gamma)\boldsymbol{e}_y$ and $\boldsymbol{\omega}_3 = \boldsymbol{\omega}_5 = R_x(\gamma)\boldsymbol{e}_z$.

Joint origins. Starting at joint 1, each origin was found by adding the relevant robot dimensions to the preceding origin. Each offset must first be expressed in the base frame. With lengths in metres, this gives

Table 2: Home screw data in `base_link`; vectors are rows, lengths in metres.

i	$\boldsymbol{\omega}_i^\mathsf{T}$	$\boldsymbol{r}_i^\mathsf{T}$	$\boldsymbol{v}_i^\mathsf{T}$
1	$(0,0,1)$	$(0,0,\frac{1}{2})$	$(0,0,0)$
2	$(0,\frac{\sqrt{3}}{2},\frac{1}{2})$	$(\frac{4}{5},0,1)$	$(-\frac{\sqrt{3}}{2},-\frac{2}{5},\frac{2\sqrt{3}}{5})$
3	$(0,-\frac{1}{2},\frac{\sqrt{3}}{2})$	$(\frac{14}{5},\frac{5\sqrt{3}}{8},\frac{13}{8})$	$(\frac{7}{4},-\frac{7\sqrt{3}}{5},-\frac{7}{5})$
4	$(1,0,0)$	$(\frac{43}{10},\frac{5\sqrt{3}-3}{8},\frac{13+3\sqrt{3}}{8})$	$(0,\frac{13+3\sqrt{3}}{8},\frac{3-5\sqrt{3}}{8})$
5	$(0,-\frac{1}{2},\frac{\sqrt{3}}{2})$	$(\frac{53}{10},\frac{5\sqrt{3}-3}{8},\frac{13+3\sqrt{3}}{8})$	$(\frac{7}{4},-\frac{53\sqrt{3}}{20},-\frac{53}{20})$
6	$(1,0,0)$	$(\frac{34}{5},\frac{5\sqrt{3}-3}{8},\frac{13+3\sqrt{3}}{8})$	$(0,\frac{13+3\sqrt{3}}{8},\frac{3-5\sqrt{3}}{8})$

$$\begin{aligned} \boldsymbol{r}_1 &= \tfrac{1}{2}\boldsymbol{e}_z, \qquad \boldsymbol{r}_2 = \boldsymbol{r}_1 + \tfrac{4}{5}\boldsymbol{e}_x + \tfrac{1}{2}\boldsymbol{e}_z, \qquad \boldsymbol{r}_3 = \boldsymbol{r}_2 + 2\boldsymbol{e}_x + \tfrac{5}{4}\boldsymbol{\omega}_2, \\ \boldsymbol{r}_4 &= \boldsymbol{r}_3 + \tfrac{3}{2}\boldsymbol{e}_x + \tfrac{3}{4}\boldsymbol{\omega}_3, \qquad \boldsymbol{r}_5 = \boldsymbol{r}_4 + \boldsymbol{e}_x, \qquad \boldsymbol{r}_6 = \boldsymbol{r}_5 + \tfrac{3}{2}\boldsymbol{e}_x. \end{aligned} \tag{2}$$

For example, the offset from joint 2 to joint 3 combines a 2 m step along the base x axis and a 1.25 m step along joint 2's tilted axis. The resulting vectors are listed in Table 2, together with $\boldsymbol{v}_i = -\boldsymbol{\omega}_i \times \boldsymbol{r}_i$.

Forward model. Each pair $(\boldsymbol{\omega}_i, \boldsymbol{v}_i)$ defines a screw axis and its matrix form:

$$\boldsymbol{\xi}_i = \begin{pmatrix} \boldsymbol{\omega}_i \\ \boldsymbol{v}_i \end{pmatrix}, \qquad \widehat{\boldsymbol{\xi}}_i = \begin{pmatrix} [\boldsymbol{\omega}_i]_\times & \boldsymbol{v}_i \\ \mathbf{0}^\mathsf{T} & 0 \end{pmatrix}, \qquad [\boldsymbol{\omega}_i]_\times \boldsymbol{p} = \boldsymbol{\omega}_i \times \boldsymbol{p}. \tag{3}$$

The endpoint has orientation $R_x(\gamma)$ and position $\boldsymbol{r}_6$ at home. This gives its home pose M and the complete forward kinematics:

$$M = \begin{pmatrix} R_x(\gamma) & \boldsymbol{r}_6 \\ \mathbf{0}^\mathsf{T} & 1 \end{pmatrix}, \qquad \boxed{{}^0T_E(\boldsymbol{q}) = e^{\widehat{\boldsymbol{\xi}}_1 q_1} e^{\widehat{\boldsymbol{\xi}}_2 q_2} \cdots e^{\widehat{\boldsymbol{\xi}}_6 q_6} M.} \tag{4}$$

1.3 Verification

The DH and PoE models were compared at 514 configurations: home, each joint alone at $\pm\pi/2$, one mixed pose, and 500 uniformly sampled configurations in $[-\pi,\pi]^6$ (random seed 42). At each configuration, we computed the full pose Frobenius residual $\|\Delta T\|_F = \sqrt{\|\Delta \boldsymbol{p}\|_2^2 + \|\Delta R\|_F^2}$, where Δ denotes the difference between the models, $\|\Delta \boldsymbol{p}\|_2$ is the Euclidean position residual (positions in metres), and $\|\Delta R\|_F$ is the Frobenius rotation-matrix residual. The maximum full pose residual was below 10^{-14}, consistent with floating-point roundoff.

2 Jacobian and singularities

In this section we construct the geometric Jacobian of CuRo6R from the PoE model. The MATLAB PoE code keeps the intermediate transforms. These give each joint's current

unit direction $\boldsymbol{w}_i$ and a point $\boldsymbol{p}_i$ on its axis, both in `base_link`. The current axis direction $\boldsymbol{w}_i(\boldsymbol{q})$ is obtained by rotating the home direction $\boldsymbol{\omega}_i$ through the preceding joint motions. For a revolute joint, the Jacobian column is

$$ {}^0J_k(:,i) = \begin{pmatrix} \boldsymbol{w}_i \times (\boldsymbol{p}_k - \boldsymbol{p}_i) \\ \boldsymbol{w}_i \end{pmatrix}, \qquad i = 1, \ldots, 6. \tag{5} $$

The upper rows give linear velocity and the lower rows give angular velocity. We use $k = E$ for the endpoint. Choosing $k = 5$ instead refers the same tool motion to the wrist centre.

Arm and wrist blocks. Here ${}^3J_5$ denotes this full Jacobian at the wrist centre, expressed along the physical link-3 axes (`link_3_d3`). Both velocity blocks are rotated by $({}^0R_3)^T$; the velocities remain relative to the base. All three wrist axes pass through $\boldsymbol{p}_5$, so their linear columns vanish. With $c_i = \cos q_i$ and $s_i = \sin q_i$, the result has the form

$$ {}^3J_5 = \begin{pmatrix} J_{v,\mathrm{arm}} & 0 \\ J_{\omega,\mathrm{arm}} & J_{\omega,\mathrm{wrist}} \end{pmatrix}, \qquad J_{\omega,\mathrm{wrist}} = \begin{pmatrix} 1 & 0 & c_5 \\ 0 & -s_4 & c_4 s_5 \\ 0 & c_4 & s_4 s_5 \end{pmatrix}. \tag{6} $$

The subscripts v and ω denote linear and angular velocity. The 3×3 blocks $J_{v,\mathrm{arm}}$ and $J_{\omega,\mathrm{arm}}$ map the rates of joints 1–3 to these two velocities; $J_{\omega,\mathrm{wrist}}$ maps the rates of joints 4–6 to angular velocity. Their linear contribution at the wrist centre is zero.

We obtain an exact symbolic expression of ${}^3J_5$ in MATLAB, from which the determinant is extracted. The zero block of $J = {}^3J_5$ makes the determinant easy to evaluate:

$$ \det J(\boldsymbol{q}) = -\sin q_5 \, \det J_{v,\mathrm{arm}}(q_2, q_3). \tag{7} $$

Using the DH lengths, define $F_1 = a_2 + d_4 c_3$, $X = F_1 c_2 + d_3 s_2$, and $D = -F_1 s_2 + d_3 c_2$. The explicit arm determinant is

$$ \det J_{v,\mathrm{arm}} = -\frac{d_4}{2} \left[a_1 D c_3 - \sqrt{3}\, a_1 F_1 s_3 - \sqrt{3}\, X (a_2 s_3 - d_2 c_3) \right]. \tag{8} $$

Changing the expression frame or velocity reference point preserves $\det J$, provided all six columns describe the same full tool motion. The wrist factor vanishes at $q_5 = 0$ or π modulo 2π. Axes 4 and 6 then lie on the same line, so one independent rotation is lost. The arm factor depends jointly on q_2 and q_3. Thus the wrist and arm conditions separate, while the arm condition must be studied in the (q_2, q_3) plane, as shown in Figure 1(a) in Section 4.

A full configuration is regular exactly when both factors are nonzero. For numerical path validation in Section 4, we monitor $\sigma_{\min}(\widetilde{J})$, using the scaled endpoint Jacobian defined there. It vanishes at singular configurations; a positive threshold enforces a numerical margin.

3 Analytical inverse kinematics

3.1 Frames, geometry and the wrist target

We seek the joint vectors $\boldsymbol{q} = (q_1, \ldots, q_6)^\mathsf{T}$ giving a prescribed pose of the endpoint frame E (`link_6`). Frame 0 is `base_link`; frames $1, \ldots, 5$ are the physical URDF link frames, as in Section 2. The q_i are URDF joint displacements from home. We use the DH lengths from Table 1 and the fixed tilt γ from Subsection 1.2.

Let W be the wrist centre, the origin of frame 5. The geometry is specified by

$$\begin{aligned} {}^0\boldsymbol{t}_2\big|_{q_1=0} &= (a_1, 0, d_1)^\mathsf{T}, & {}^2\boldsymbol{t}_3 &= (a_2, d_2, 0)^\mathsf{T}, \\ {}^3\boldsymbol{p}_W &= (d_4, 0, d_3)^\mathsf{T}, & Q &:= {}^1R_2\big|_{q_2=0} = R_x(\gamma). \end{aligned} \tag{9}$$

We retain $c_i = \cos q_i$, $s_i = \sin q_i$ and define $c_\gamma = \cos\gamma$, $s_\gamma = \sin\gamma$. The matrices R_x, R_y, R_z are right-handed rotations.

All three wrist axes pass through W, so its position depends only on q_1, q_2, q_3. The origin of E lies a distance d_6 from W along its own x axis. The subscript d denotes a desired quantity, so the target pose gives the desired wrist position

$$\boxed{{}^0\boldsymbol{p}_{W,d} = {}^0\boldsymbol{t}_{E,d} - d_6\,{}^0R_{E,d}\boldsymbol{e}_x = \begin{pmatrix} x_{w,d} & y_{w,d} & z_{w,d} \end{pmatrix}^\mathsf{T}}. \tag{10}$$

Wrist-centre position. Let O be the point on axis 1 at shoulder height, ${}^0\boldsymbol{p}_O = d_1\boldsymbol{e}_z$. Combining the relative translations and rotations gives

$${}^0\boldsymbol{p}_W(\boldsymbol{q}) = \underbrace{\underbrace{d_1\boldsymbol{e}_z}_{{}^0\boldsymbol{p}_O} + \underbrace{R_z(q_1)}_{{}^0R_1}\Bigg[\underbrace{a_1\boldsymbol{e}_x}_{{}^1\boldsymbol{t}_2} - {}^1\boldsymbol{p}_O + \underbrace{QR_y(q_2)}_{{}^1R_2}\Bigg(\underbrace{\begin{pmatrix} a_2 \\ d_2 \\ 0 \end{pmatrix}}_{{}^2\boldsymbol{t}_3} + \underbrace{R_z(q_3)}_{{}^2R_3}\underbrace{\begin{pmatrix} d_4 \\ 0 \\ d_3 \end{pmatrix}}_{{}^3\boldsymbol{p}_W}\Bigg)}_{{}^2\boldsymbol{p}_W}\Bigg]}_{{}^1\boldsymbol{p}_W - {}^1\boldsymbol{p}_O}. \tag{11}$$

This is the wrist-centre position obtained from the PoE model. We evaluate it from the inside out. First, joint 3 gives the shoulder-to-wrist coordinates in frame 2: ${}^2\boldsymbol{p}_W = (F_1, B, F_2)^\mathsf{T}$, where

$$F_1 := a_2 + d_4c_3, \qquad B := d_2 + d_4s_3, \qquad F_2 := d_3. \tag{12}$$

Joint 2 leaves B unchanged. Applying this rotation, then the fixed tilt and shoulder offset, gives ${}^1\boldsymbol{p}_W - {}^1\boldsymbol{p}_O = (A, Y, H)^\mathsf{T}$

$$\begin{aligned} X &= F_1c_2 + F_2s_2, & D &= F_2c_2 - F_1s_2, \\ A &= a_1 + X, & Y &= c_\gamma B - s_\gamma D, & H &= s_\gamma B + c_\gamma D. \end{aligned} \tag{13}$$

For an IK solution, the final base rotation gives

$$x_{w,d} = Ac_1 - Ys_1, \qquad y_{w,d} = As_1 + Yc_1, \qquad z_{w,d} = d_1 + H. \tag{14}$$

3.2 Eliminating the first two joint angles

We now form three *known target quantities*:

$$h := z_{w,d} - d_1, \qquad \rho_{w,d}^2 := x_{w,d}^2 + y_{w,d}^2, \qquad \mathcal{R} := \rho_{w,d}^2 + h^2 = \|{}^0\boldsymbol{p}_{W,d} - {}^0\boldsymbol{p}_O\|^2. \tag{15}$$

h is the desired wrist height above O, $\rho_{w,d}$ is its distance to axis 1, and $\mathcal{R}$ is the wrist squared distance to O. None changes when joint 1 rotates. For a solution, (14) gives $H = h$ and $\rho_{w,d}^2 = A^2 + Y^2$.

The q_2 rotation in (13) preserves the norm:

$$\underbrace{X^2 + B^2 + D^2}_{\|R_y(q_2)(F_1,B,F_2)^\mathsf{T}\|^2} = \underbrace{F_1^2 + B^2 + F_2^2}_{\|(F_1,B,F_2)^\mathsf{T}\|^2} \quad\Longrightarrow\quad X^2 + D^2 = F_1^2 + F_2^2. \tag{16}$$

The unchanged y component B cancels. Likewise, $(Y, H)^\mathsf{T} = \left(\begin{smallmatrix} c_\gamma & -s_\gamma \\ s_\gamma & c_\gamma \end{smallmatrix}\right)(B, D)^\mathsf{T}$ gives $Y^2 + h^2 = B^2 + D^2$. Hence

$$\begin{aligned} \mathcal{R} &= (a_1 + X)^2 + B^2 + D^2 = 2a_1X + F_3, \qquad h = s_\gamma B + c_\gamma D, \\ F_3 &:= a_1^2 + F_1^2 + F_2^2 + B^2 = L + 2a_2d_4c_3 + 2d_2d_4s_3, \\ L &:= a_1^2 + a_2^2 + d_2^2 + d_3^2 + d_4^2. \end{aligned} \tag{17}$$

The expansion of F_3 uses $c_3^2 + s_3^2 = 1$: F_3 depends only on q_3, while L is constant. With $\mathcal{R}, h$ known and a_1, c_γ nonzero, a trial q_3 fixes

$$X = \frac{\mathcal{R} - F_3}{2a_1}, \qquad D = \frac{h - s_\gamma B}{c_\gamma}. \tag{18}$$

These required X, D can come from a real rotation q_2 only if their norm is correct. Substituting into (16) and multiplying by $4a_1^2$ gives us the *closure condition*:

$$\boxed{G(q_3) := (\mathcal{R} - F_3)^2 + \frac{4a_1^2}{c_\gamma^2}(h - s_\gamma B)^2 - 4a_1^2(F_1^2 + d_3^2) = 0.} \tag{19}$$

Only q_3 remains unknown. For $\gamma = 0$, $D = h$ and this becomes the lecture's untilted closure, with terminal arm length d_4.

3.3 Converting the closure into a quartic

We use the half-angle substitution

$$t = \tan(q_3/2), \qquad \Delta = 1 + t^2, \qquad c_3 = \frac{1 - t^2}{\Delta}, \qquad s_3 = \frac{2t}{\Delta}. \tag{20}$$

Substituting into (19) and multiplying by $c_\gamma^2\Delta^2$ clears the denominators and gives

$$\boxed{\begin{aligned} P(t) = c_\gamma^2\Big[(\mathcal{R} - L)\Delta - 2a_2d_4(1 - t^2) - 4d_2d_4t\Big]^2 \\ + 4a_1^2\Big[h\Delta - s_\gamma(d_2\Delta + 2d_4t)\Big]^2 - 4a_1^2c_\gamma^2\Big\{[a_2\Delta + d_4(1 - t^2)]^2 + d_3^2\Delta^2\Big\} = 0. \end{aligned}} \tag{21}$$

The expressions being squared have degree at most two, so P has degree at most four. Every real root gives a $q_3 = 2\arctan t$. Since $\Delta > 0$ for real t, clearing denominators adds no real roots. We must test $q_3 = \pi$ separately in (19), because its half-angle tangent is undefined. This missing point is included in the maximum of four distinct q_3 candidates.

3.4 Recovering and checking the solutions

Arm angles. For each candidate q_3, we evaluate (12) for F_1, F_2, B and use (18) for X, D. Combining their defining equations cancels s_2 in $F_1X + F_2D$ and c_2 in $F_2X - F_1D$, giving us

$$c_2 = \frac{F_1X + F_2D}{F_1^2 + F_2^2}, \qquad s_2 = \frac{F_2X - F_1D}{F_1^2 + F_2^2}, \qquad q_2 = \operatorname{atan2}(s_2, c_2). \tag{22}$$

The denominator is positive because $F_2 = d_3 \neq 0$, and the closure ensures $c_2^2 + s_2^2 = 1$. Next we set $A = a_1 + X$, $Y = c_\gamma B - s_\gamma D$ and recover the base angle from the horizontal equations:

$$q_1 = \operatorname{atan2}(\underbrace{Ay_{w,d} - Yx_{w,d}}_{\rho_{w,d}^2 s_1}, \underbrace{Ax_{w,d} + Yy_{w,d}}_{\rho_{w,d}^2 c_1}), \qquad \rho_{w,d} > 0. \tag{23}$$

The common positive factor $\rho_{w,d}^2$ does not change the angle. If $\rho_{w,d} = 0$, the wrist centre lies on axis 1 and q_1 is free, in which case we choose a value in $[-\pi, \pi]$ and solve the wrist for that choice.

Wrist angles. For each arm solution, we undo its orientation using the transpose, then solve the remaining local x–z–x wrist rotations:

$$\begin{aligned} {}^0R_3 &= R_z(q_1)QR_y(q_2)R_z(q_3), \\ {}^3R_{E,d} &= ({}^0R_3)^\mathsf{T}\,{}^0R_{E,d} = R_x(q_4)R_z(q_5)R_x(q_6) =: (r_{ij}). \end{aligned} \tag{24}$$

Here r_{ij} is entry (i, j) of ${}^3R_{E,d}$. Multiplying the wrist rotations gives us five useful entries:

$$r_{11} = c_5, \quad r_{21} = c_4 s_5, \quad r_{31} = s_4 s_5, \quad r_{12} = -s_5 c_6, \quad r_{13} = s_5 s_6. \tag{25}$$

Let $\eta = \sqrt{r_{21}^2 + r_{31}^2} = |s_5|$. For $\eta > 0$, we choose each sign $\varepsilon \in \{-1, +1\}$, so that $s_5 = \varepsilon\eta$. Then

$$\boxed{\begin{aligned} &q_5 = \operatorname{atan2}(\varepsilon\eta, r_{11}), \\ &q_4 = \operatorname{atan2}(\varepsilon r_{31}, \varepsilon r_{21}), \qquad q_6 = \operatorname{atan2}(\varepsilon r_{13}, -\varepsilon r_{12}). \end{aligned}} \tag{26}$$

We must use the same sign in all three formulas. It selects one of two wrist configurations with the same orientation. If $\eta = 0$, joints 4 and 6 are coaxial, so only a sum or difference is fixed:

$$\begin{aligned} r_{11} = 1: &\quad q_5 = 0, \quad q_4 + q_6 = \operatorname{atan2}(r_{32}, r_{22}), \\ r_{11} = -1: &\quad q_5 = \pi, \quad q_6 - q_4 = \operatorname{atan2}(r_{32}, -r_{22}). \end{aligned} \tag{27}$$

We may set $q_4 = 0$ and recover q_6 from the appropriate relation.

Assembly and validation. For every real q_3 candidate, we recover the arm and both wrist branches. We represent angles in $[-\pi, \pi]$, enforce joint limits, remove duplicates modulo 2π, and check position and orientation by forward kinematics. We use numerical tolerances for roots, singularities and pose errors. Away from continuous singular families, at most four arm solutions each have two wrist branches, so there are **at most eight full-pose solutions**.

4 Connectivity analysis

In this section we provide a qualitative analysis of the conditions required for a non-singular change of solutions, and the sequence of steps we used to numerically get a trajectory for a nonsingular change of solution with constant tool orientation.

A straight arm path with fixed tool orientation. We index each full configuration $\boldsymbol{q}^{(i)}$ by its arm root t_i on the chosen wrist branch. We choose the initial joint configuration $\boldsymbol{q}^{(4)} = (0.1024, -1.1124, 1.8547, -1.4399, -0.3835, -2.6486)^\mathsf{T}$ (in radians), compute ${}^0T_{E,d} = \mathrm{FK}(\boldsymbol{q}^{(4)})$, and solve the IK for this pose, obtaining eight solutions. Excluding the initial solution, exactly one has the same signs of $\det J_{v,\mathrm{arm}}$ and $\sin q_5$: $\boldsymbol{q}^{(2)} \approx (-2.2287, -2.1482, -0.2788, 0.6181, -0.3524, -1.7072)^\mathsf{T}$. This candidate also satisfies the endpoint joint-limit and singularity margins, but matching signs alone does not guarantee a non-singular connection.

We discretize the straight path between their arm joints $\boldsymbol{q}_a = (q_1, q_2, q_3)^\mathsf{T}$:

$$\boldsymbol{q}_a(s) = (1 - s)\boldsymbol{q}_a^{(4)} + s\boldsymbol{q}_a^{(2)}, \qquad 0 \le s \le 1, \tag{28}$$

where s measures path advancement. At each sample, we remove the arm orientation from the desired orientation ${}^0R_{E,d}$ by computing ${}^3R_{E,d}(s) = ({}^0R_3(\boldsymbol{q}_a(s)))^\mathsf{T}\,{}^0R_{E,d}$. We recover

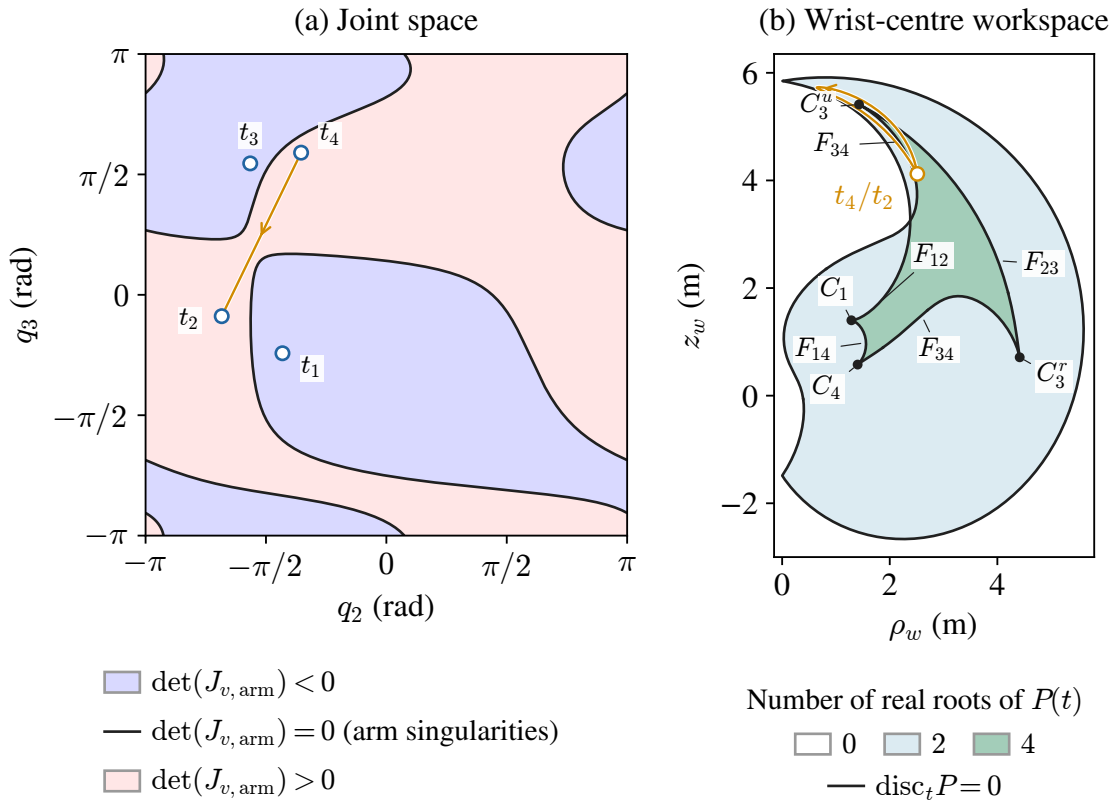


Figure 1: Arm singularities and a fixed-orientation solution change in (a) joint space and (b) the wrist-centre workspace, with $\rho_w = \sqrt{x_w^2 + y_w^2}$. The cusps indicate that the CuRo6R's positioning chain is cuspidal: it can change between IK solutions without crossing an arm singularity. At the marked target, $t_1 < t_2 < t_3 < t_4$. The folds F_{ij} merge t_i, t_j; cusps C_k are indexed by the root k shared by their incident folds, with u/r superscript distinguishing upper/right.

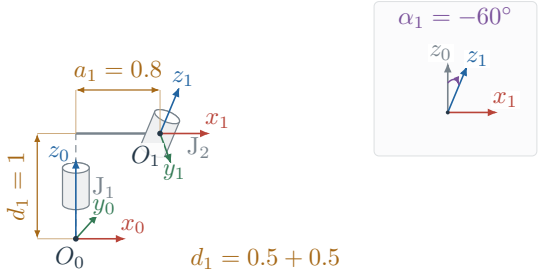
the wrist angles using (26), preserving the initial sign of $\sin q_5$ and choosing continuous angle representatives. The tool orientation remains fixed while its position is free to vary. We continue only while joint-limit clearance is nonnegative, the arm determinant retains its sign, $|\sin q_5| > 10^{-6}$, and $\sigma_{\min}(\widetilde{J}) > 10^{-3}$, where $\widetilde{J} = \mathrm{diag}(\ell_c^{-1} I_3, I_3)\, {}^0J_E$ uses the characteristic length $\ell_c = 2\,\mathrm{m}$ to scale the linear velocity rows. A failed check stops the attempt; acceptance also requires reaching the selected goal. Figure 1 shows the accepted change from $\boldsymbol{q}^{(4)}$ to $\boldsymbol{q}^{(2)}$. These checks concern the sampled configurations, so a finer discretization improves confidence of real success.

A DH frame constructions

Figure 2 collects all six frame-to-frame constructions at the home configuration ($\boldsymbol{q} = \mathbf{0}$), using one fixed viewing direction and length scale. Lengths are in metres; angles are shown in degrees for readability.

1 Base to shoulder: $\{0\} \to \{1\}$

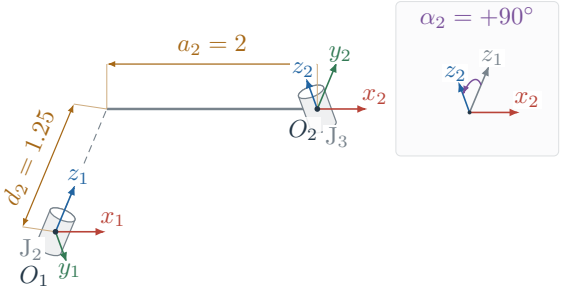
$d_1 = 1, \quad \theta_1 = q_1, \quad \alpha_1 = -60°, \quad a_1 = 0.8$



The full base height is 1 m. The 30° URDF roll of joint 2 gives $\alpha_1 = 30° - 90° = -60°$.

2 Shoulder to elbow: $\{1\} \to \{2\}$

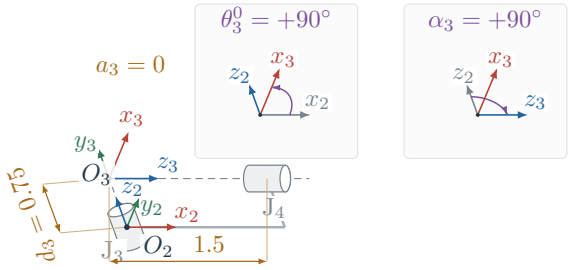
$d_2 = 1.25, \quad \theta_2 = q_2, \quad \alpha_2 = 90°, \quad a_2 = 2$



The x axes remain parallel. The offset along z_1 and the common normal along x_2 locate O_2 on joint 3.

3 Axis intersection: $\{2\} \to \{3\}$

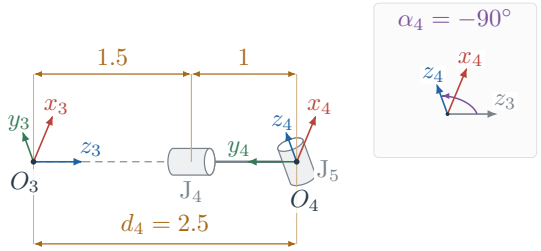
$d_3 = 0.75, \quad \theta_3 = q_3 + 90°, \quad \alpha_3 = 90°, \quad a_3 = 0$



O_3 is the intersection of the extended axes. Choosing $x_3 = y_2$ explains the fixed +90° offset in θ_3.

4 Reach the wrist: $\{3\} \to \{4\}$

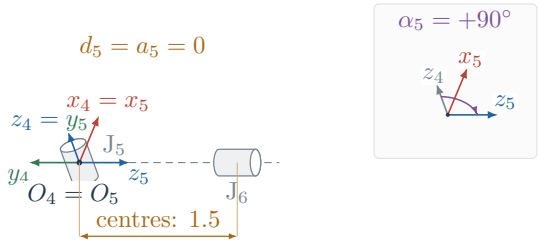
$d_4 = 2.5, \quad \theta_4 = q_4, \quad \alpha_4 = -90°, \quad a_4 = 0$



The distance starts at O_3, rather than the joint-4 cylinder. The two physical offsets add: $d_4 = 1.5 + 1$.

5 Coincident origins: $\{4\} \to \{5\}$

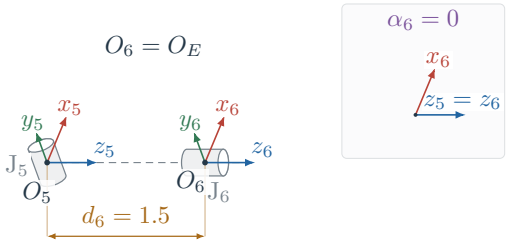
$d_5 = 0, \quad \theta_5 = q_5, \quad \alpha_5 = 90°, \quad a_5 = 0$



The origins coincide, although the joint centres do not. Here $x_4 = x_5$ and the extended joint-6 axis passes through the wrist centre.

6 Wrist to endpoint: $\{5\} \to \{6\}$

$d_6 = 1.5, \quad \theta_6 = q_6, \quad \alpha_6 = 0, \quad a_6 = 0$



At home the two frames have parallel axes. $O_6 = O_E$; the fixed endpoint alignment changes orientation only.

Figure 2: Standard DH frame assignment. Cylinders denote physical revolute joints J_i; dots mark DH origins O_i. x, y and z axes retain their colours. Dashed lines extend joint axes; gold dimensions locate the origins. Boxed angle sketches show the rotation $x_2 \to x_3$ about $+z_2$, giving θ_3^0, and the signed twist $z_{i-1} \to z_i$ about $+x_i$ (with the first direction translated to the same point). Cylinder orientation, size and viewing direction are reused between panels; drawings are schematic.